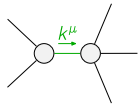
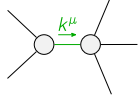


	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1\alpha\beta\chi}$						
$\mathcal{K}_{0^+}^{\#1\dagger}$	0	0						
$\mathcal{K}_{0^-}^{\#1\dagger\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{\#1\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\alpha\beta}$	$\mathcal{K}_{1^-}^{\#1\alpha}$	$\mathcal{K}_{1^-}^{\#2\alpha}$		
	$\mathcal{K}_{1^+}^{\#1\dagger\alpha\beta}$	$-k^2 \frac{(4)}{K_3}$	$\frac{k^2 (4)}{2 \sqrt{2}} \frac{K_4}{2}$	0	0			
	$\mathcal{K}_{1^+}^{\#2\dagger\alpha\beta}$	$\frac{k^2 (4)}{2 \sqrt{2}} \frac{K_4}{2}$	$\frac{Y_1 k^2}{2}$	0	0			
	$\mathcal{K}_{1^+}^{\#1\dagger\alpha}$	0	0	$k^2 \frac{(4)}{K_1}$	$\sqrt{2} k^2 \frac{(4)}{K_1}$			
	$\mathcal{K}_{1^+}^{\#2\dagger\alpha}$	0	0	$\sqrt{2} k^2 \frac{(4)}{K_1}$	$2 k^2 \frac{(4)}{K_1}$	$\mathcal{K}_{2^+}^{\#1\alpha\beta}$	$\mathcal{K}_{2^-}^{\#1\alpha\beta\chi}$	
				$\mathcal{K}_{2^+}^{\#1\dagger\alpha\beta}$	0	0		
				$\mathcal{K}_{2^-}^{\#1\dagger\alpha\beta\chi}$	0	0		

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^+}^{\#1\alpha\beta\chi}$					
$\mathcal{T}_{0^+}^{\#1\dagger}$	0	0					
$\mathcal{T}_{0^+}^{\#1\dagger\alpha\beta\chi}$	0	0	$\mathcal{T}_{1^+}^{\#1\alpha\beta}$	$\mathcal{T}_{1^+}^{\#2\alpha\beta}$	$\mathcal{T}_{1^+}^{\#1\alpha}$	$\mathcal{T}_{1^+}^{\#2\alpha}$	
	$\mathcal{T}_{1^+}^{\#1\dagger\alpha\beta}$	$\frac{\Upsilon_1 k^2}{2 \text{Det}(1^+)}$	$-\frac{k^2 \binom{4}{\kappa_4}}{2 \sqrt{2} \text{Det}(1^+)}$	0	0		
	$\mathcal{T}_{1^+}^{\#2\dagger\alpha\beta}$	$-\frac{k^2 \binom{4}{\kappa_4}}{2 \sqrt{2} \text{Det}(1^+)}$	$-\frac{k^2 \binom{4}{\kappa_3}}{\text{Det}(1^+)}$	0	0		
	$\mathcal{T}_{1^+}^{\#1\dagger\alpha}$	0	0	$\frac{1}{9 k^2 \binom{4}{\kappa_1}}$	$\frac{\sqrt{2}}{9 k^2 \binom{4}{\kappa_1}}$		
	$\mathcal{T}_{1^+}^{\#2\dagger\alpha}$	0	0	$\frac{\sqrt{2}}{9 k^2 \binom{4}{\kappa_1}}$	$\frac{2}{9 k^2 \binom{4}{\kappa_1}}$	$\mathcal{T}_{2^+}^{\#1\alpha\beta}$	$\mathcal{T}_{2^+}^{\#1\alpha\beta\chi}$
				$\mathcal{T}_{2^+}^{\#1\dagger\alpha\beta}$	0	0	
				$\mathcal{T}_{2^+}^{\#1\dagger\alpha\beta\chi}$	0	0	

Abbreviations used in matrices			
$Y_1 = 9 \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4} \&\& \text{Det}(1^+) = -\frac{1}{8} k^4 (36 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_3} + (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2)$			
Lagrangian			
$\begin{aligned} & \binom{(4)}{\kappa_1} \partial_\beta \mathcal{K}_{\chi^\delta}^\delta \partial^\chi \mathcal{K}_\alpha^\alpha - \binom{(4)}{\kappa_1} \partial_\chi \mathcal{K}_{\beta^\delta}^\delta \partial^\chi \mathcal{K}_\alpha^\alpha + \binom{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \binom{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \binom{(4)}{\kappa_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + \\ & 6 \binom{(4)}{\kappa_1} \partial^\chi \mathcal{K}_\alpha^\alpha \partial_\delta \mathcal{K}_{\chi\beta}^\delta + \frac{9}{2} \binom{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{1}{2} \binom{(4)}{\kappa_3} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{1}{2} \binom{(4)}{\kappa_4} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta \end{aligned}$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} = 0$	1	$\begin{aligned} & \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} = \\ & \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} \end{aligned}$	
$\mathcal{J}_{0^+}^{\#1} = 0$	1	$\partial_\beta \mathcal{J}^\alpha{}_\alpha = 0$	
$\mathcal{J}_1^{\#1\alpha} = \mathcal{J}_1^{\#2\alpha}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta = 0$	
$\mathcal{J}_2^{\#1\alpha\beta\chi} = 0$	5	$\begin{aligned} & 3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \\ & 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}{}_\delta{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\delta \mathcal{J}^{\delta\alpha}{}_\delta = \\ & 3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \\ & 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}{}_\delta{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\delta \mathcal{J}^{\delta\beta}{}_\delta \end{aligned}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} = 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial_\delta \mathcal{J}^\chi{}_\chi{}^\delta = 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^\chi{}_\chi{}^\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	15		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$-\frac{48 \binom{(4)}{\kappa_1}^2 + 8 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2}{\binom{(4)}{\kappa_1} (36 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_3} + (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2)}$
	1	0	$-\frac{216 \binom{(4)}{\kappa_1}^2 + 24 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) + (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2}{\binom{(4)}{\kappa_1} (36 \binom{(4)}{\kappa_1} \binom{(4)}{\kappa_3} + (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2)}$

$$\begin{aligned}
& \left(\kappa_4^{(4)} < 0 \& \& \left(\left(\kappa_3^{(4)} < 0 \& \& \left(\kappa_1^{(4)} < 0 \parallel \kappa_1^{(4)} > \frac{-4 \kappa_3^{(4)2} - 4 \kappa_3^{(4)} \kappa_4^{(4)} - \kappa_4^{(4)2}}{36 \kappa_3^{(4)}} \right) \right) \parallel \left(\kappa_3^{(4)} == 0 \& \& \kappa_1^{(4)} < 0 \right) \right) \parallel \right. \\
& \left. \left(0 < \kappa_3^{(4)} < -\frac{\kappa_4^{(4)}}{2} \& \& \frac{-4 \kappa_3^{(4)2} - 4 \kappa_3^{(4)} \kappa_4^{(4)} - \kappa_4^{(4)2}}{36 \kappa_3^{(4)}} < \kappa_1^{(4)} < 0 \right) \parallel \left(\kappa_3^{(4)} > -\frac{\kappa_4^{(4)}}{2} \& \& \frac{-4 \kappa_3^{(4)2} - 4 \kappa_3^{(4)} \kappa_4^{(4)} - \kappa_4^{(4)2}}{36 \kappa_3^{(4)}} < \kappa_1^{(4)} < 0 \right) \right) \parallel \right. \\
& \left. \left(\kappa_4^{(4)} == 0 \& \& \left(\left(\kappa_3^{(4)} < 0 \& \& \left(\kappa_1^{(4)} < 0 \parallel \kappa_1^{(4)} > -\frac{\kappa_3^{(4)}}{9} \right) \right) \parallel \left(\kappa_3^{(4)} > 0 \& \& -\frac{\kappa_3^{(4)}}{9} < \kappa_1^{(4)} < 0 \right) \right) \parallel \right. \\
& \left. \left(\kappa_4^{(4)} > 0 \& \& \left(\left(\kappa_3^{(4)} < -\frac{\kappa_4^{(4)}}{2} \& \& \left(\kappa_1^{(4)} < 0 \parallel \kappa_1^{(4)} > \frac{-4 \kappa_3^{(4)2} - 4 \kappa_3^{(4)} \kappa_4^{(4)} - \kappa_4^{(4)2}}{36 \kappa_3^{(4)}} \right) \right) \parallel \right. \right. \\
& \left. \left(\kappa_3^{(4)} == -\frac{\kappa_4^{(4)}}{2} \& \& \left(\kappa_1^{(4)} < 0 \parallel \kappa_1^{(4)} > 0 \right) \right) \parallel \left(-\frac{\kappa_4^{(4)}}{2} < \kappa_3^{(4)} < 0 \& \& \left(\kappa_1^{(4)} < 0 \parallel \kappa_1^{(4)} > \frac{-4 \kappa_3^{(4)2} - 4 \kappa_3^{(4)} \kappa_4^{(4)} - \kappa_4^{(4)2}}{36 \kappa_3^{(4)}} \right) \right) \parallel \right. \\
& \left. \left(\kappa_3^{(4)} == 0 \& \& \left(\kappa_1^{(4)} < 0 \right) \parallel \left(\kappa_3^{(4)} > 0 \& \& \frac{-4 \kappa_3^{(4)2} - 4 \kappa_3^{(4)} \kappa_4^{(4)} - \kappa_4^{(4)2}}{36 \kappa_3^{(4)}} < \kappa_1^{(4)} < 0 \right) \right) \right)
\end{aligned}$$